Name: Arzoo Vaswani Batch: T23 Roll No. 125

Experiment 5

Aim - To Build the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat server

Programming in Jenkins:
Continuous Integration is a software development practice where members of a team integrate their work frequently, usually each person integrates at least daily leading to multiple integrations per day.
Each integration is verified by an automated build (including test) to detect integration errors as quickly as possible." In simple way, Continuous integration (CI) is the practice of frequently building and testing each change done to your code automatically.
Jenkins is a self-contained, open-source automation server which can be used to automate all sorts of tasks related to building, testing, and delivering or deploying software.
Our first job will execute the shell commands. The freestyle project provides enough options and features to build the complex jobs that you will need in your projects.
Example 1
Example 1.1: Deploying a freestyle app in Jenkins Creating
a job:
Naming the job and setting it as freestyle:

Selecting build type as "Execute shell":
Entering a simple command for the shell execution:
Applying and saving the project configuration:
Building the project:
Console output (after building):
Example 1.2: Taking parameters through files Contents
of script example1.cmd:
Executing script example1.cmd on the terminal:
Modifying the Jenkins project to execute the script while supplying required parameters:
Console output after building the modified project:
Example 2
Example 2.1: Running a Java program under Jenkins Creating
a simple Java program:
Compiling and running the program on the terminal:
Creating a new freestyle project:
Configure new project:
Console output after building:
Example 3
Example 3.1: Parameterise build
Creating a new freestyle project:
Enabling parameterisation and adding a String parameter:
Configuring the string parameter as Fname:
Adding a choice parameter and configuring it as City with the following choices:

Creating a script which takes 2 arguments for name and city:
Configuring build steps:
Entering parameters for build:
Console output after building:
Example 3.2: Running a Java program with parameters Creating
a Java program with an input argument:
Testing the program on the terminal:
Creating a new freestyle project:
Parameterise the project by adding a string parameter as follows:
Configure the build steps:
Entering the parameter for the build:
Console output after building:
Example 5
Example 5.1: Running a Python program Creating
a simple Python script:
Running the Python script on the terminal:
Creating a new freestyle project:
Parameterising the project with a string parameter as follows:
Configuring the build steps:
Setting the parameter for the build:
Console output after building:
Conclusion: Thus, we have successfully studied Continuous Integration and installed, configured, and understood programming with Jenkins.

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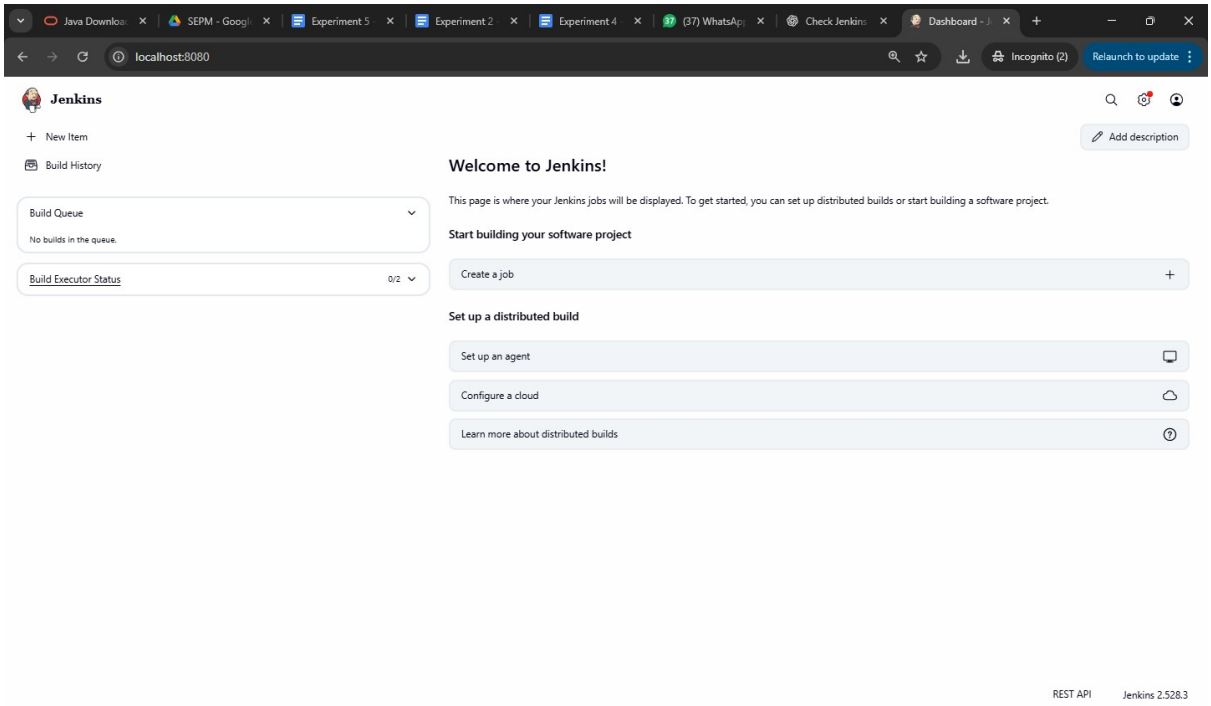

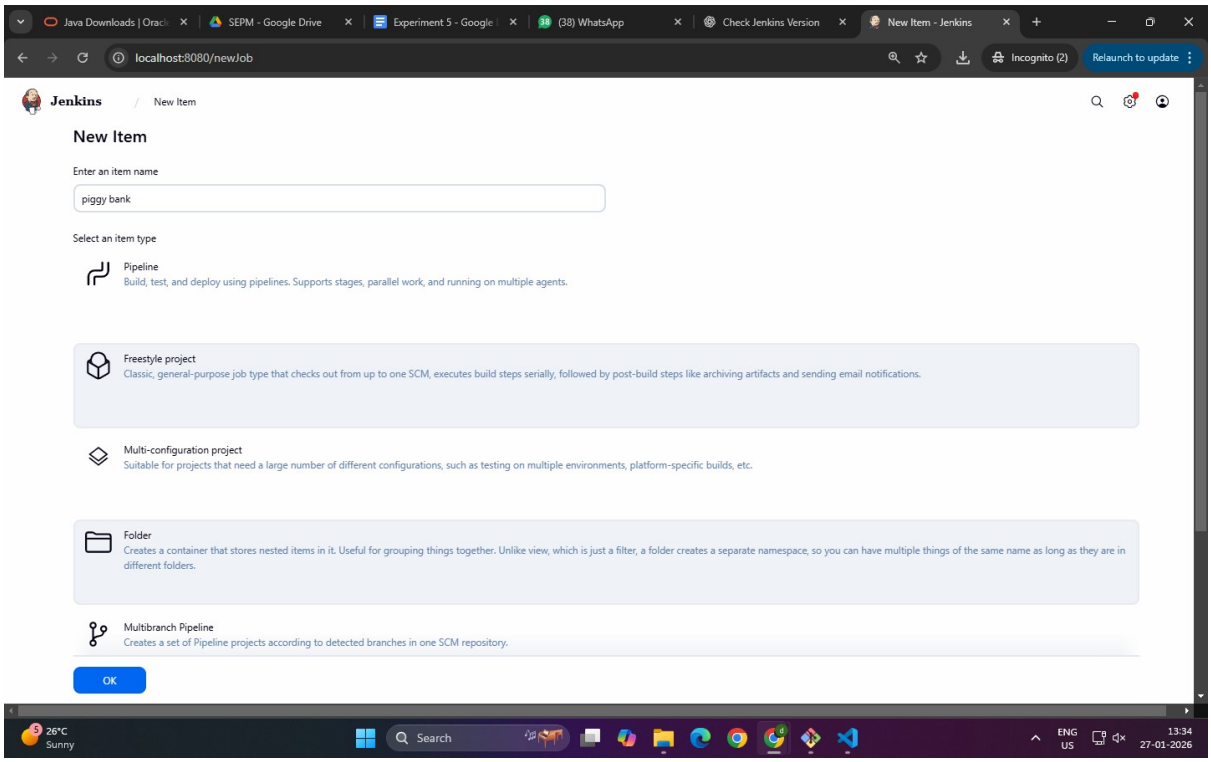

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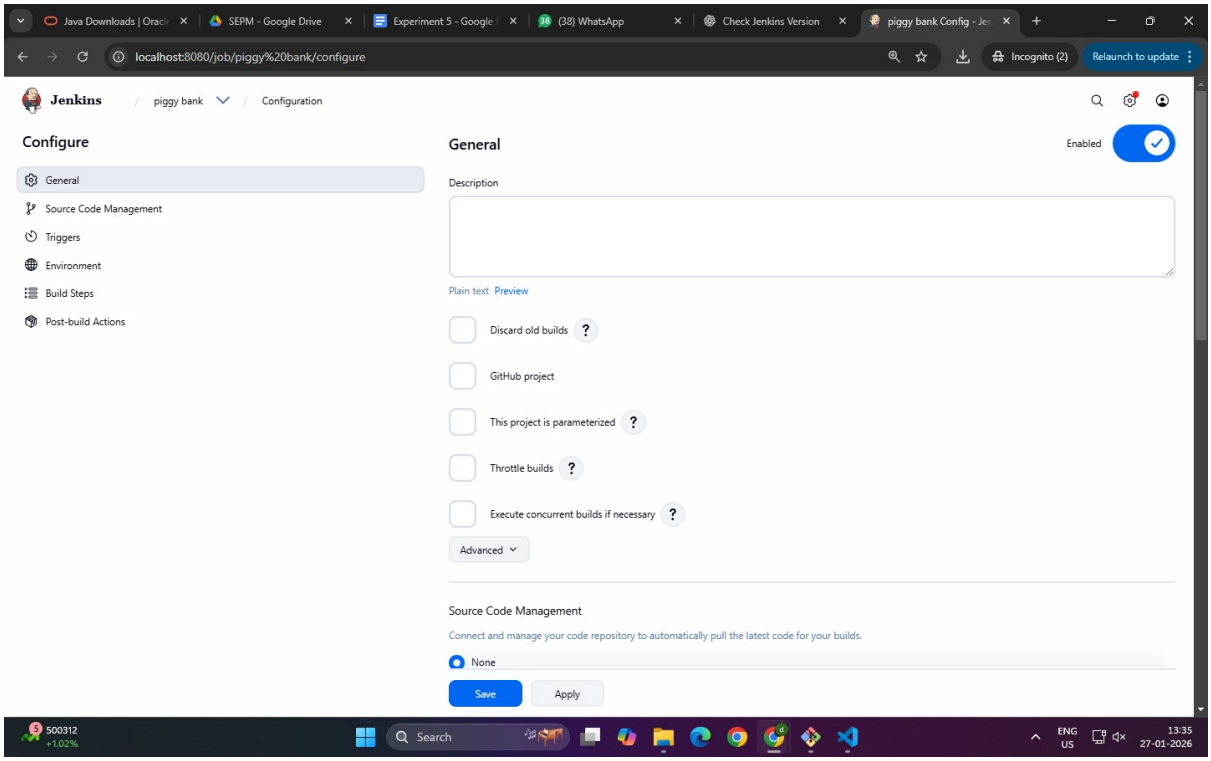

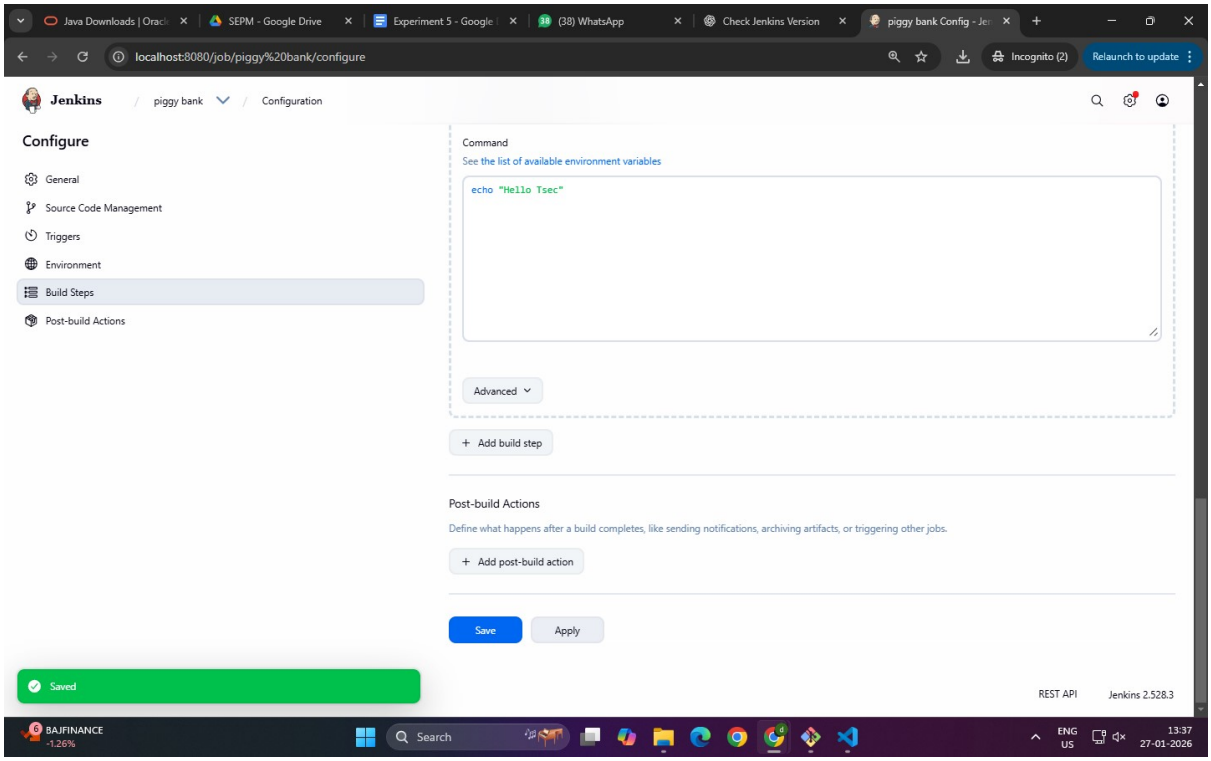

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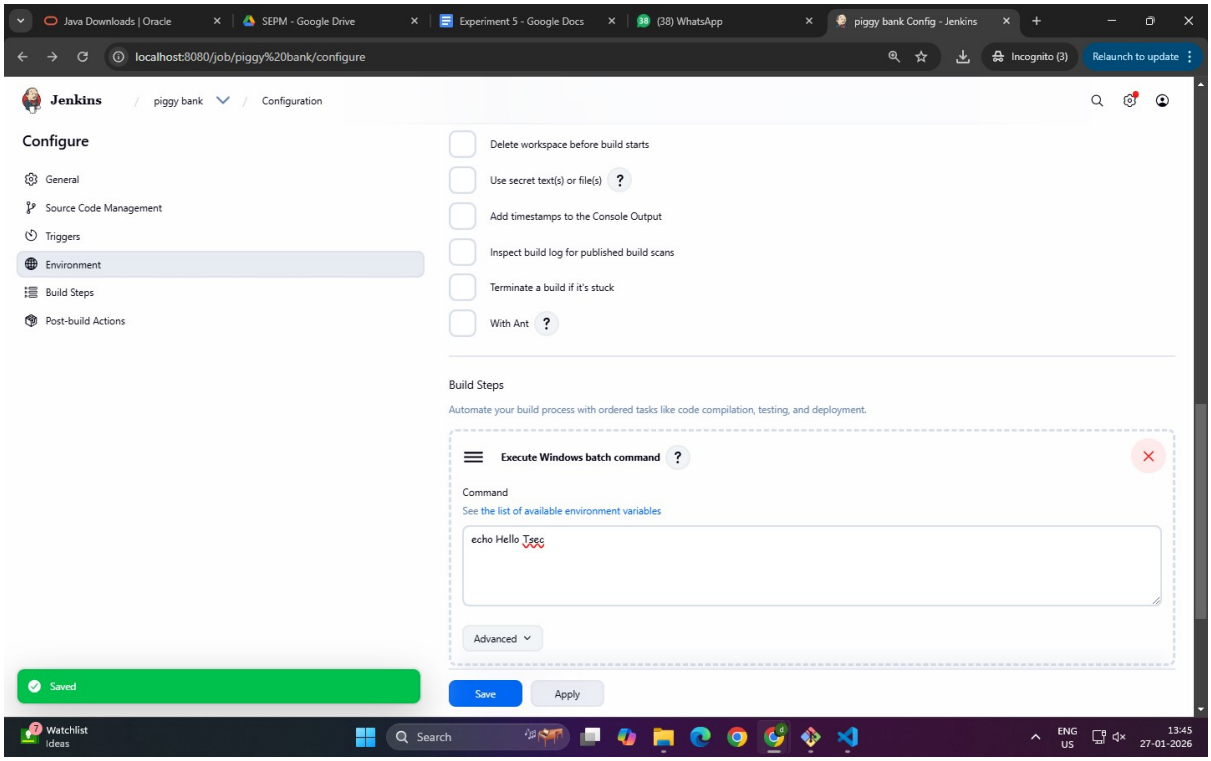

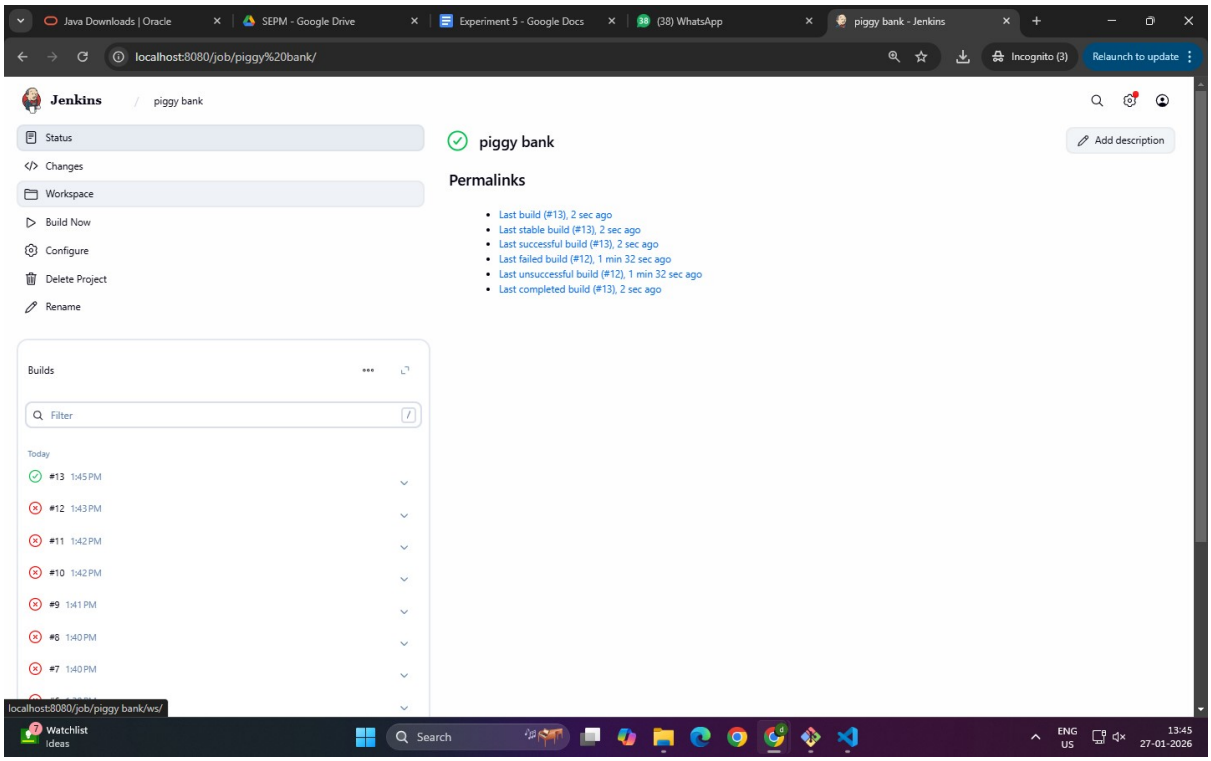

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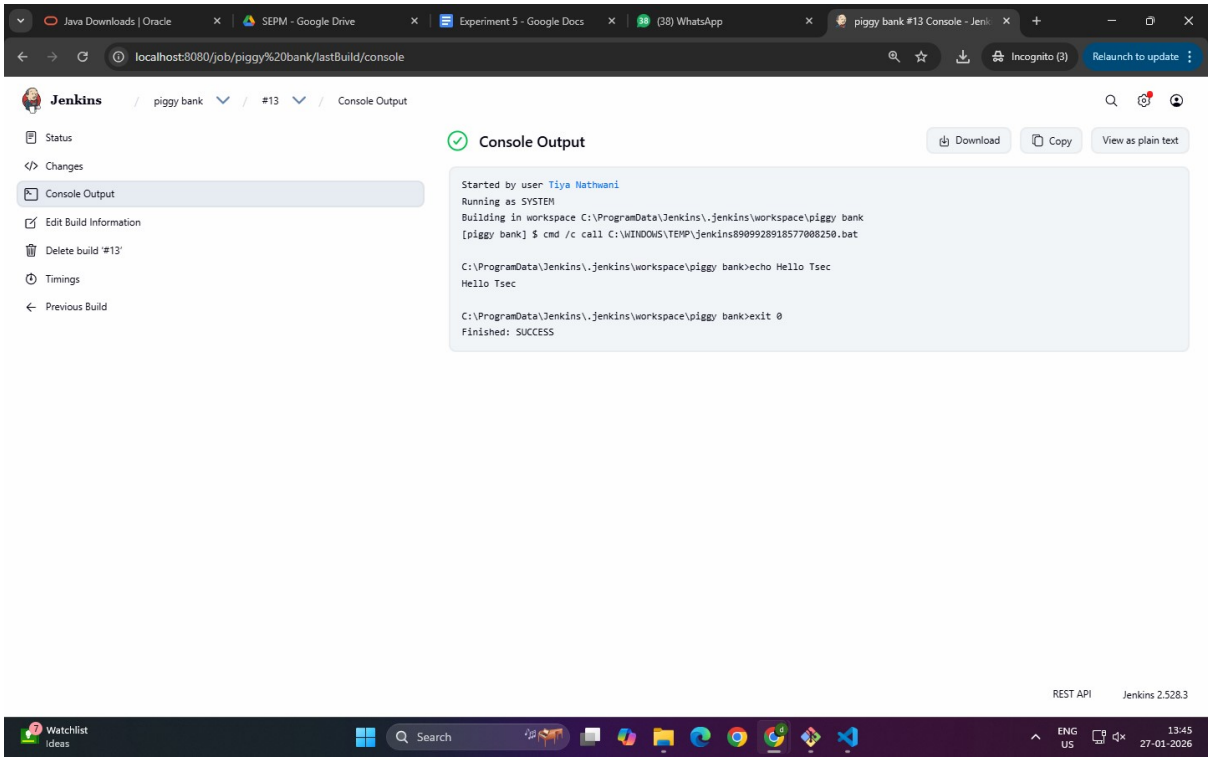

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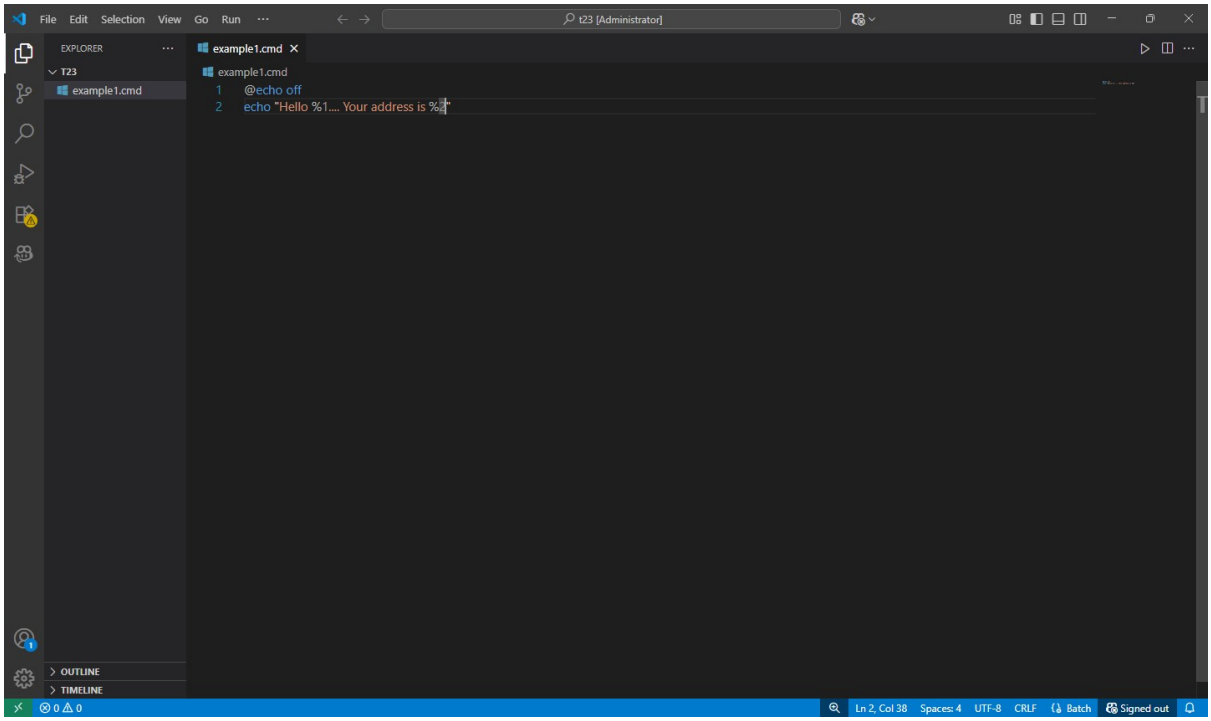
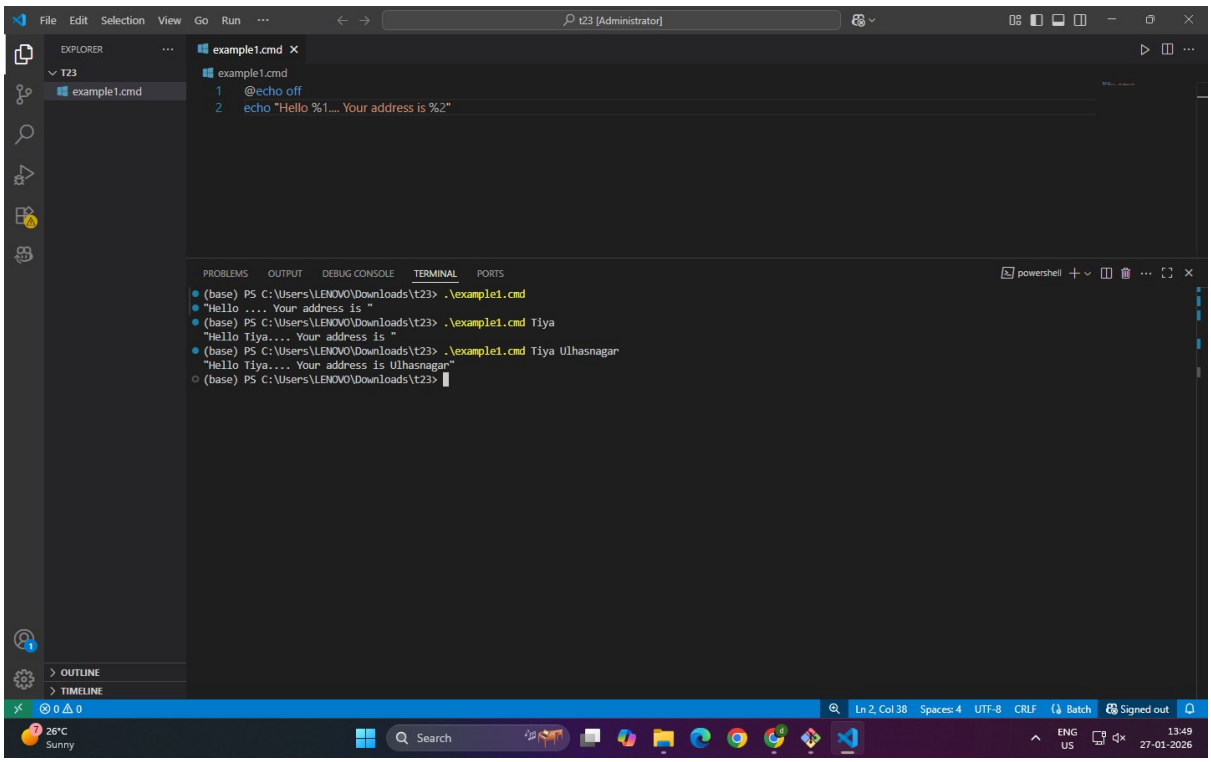

File Edit Selection View Go Run
t23 [Administrator]
EXPLORER
T23
example1.cmd
example1.cmd
@echo off
echo "Hello %1.... Your address is %2"
OUTLINE
TIMELINE
Ln 2, Col 38 Spaces: 4 UTF-8 CRLF Batch Signed out


File Edit Selection View Go Run
t23 [Administrator]
EXPLORER
T23
example1.cmd
example1.cmd
@echo off
echo "Hello %1.... Your address is %2"
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell
(base) PS C:\Users\LENOVO\Downloads\t23> .\example1.cmd
"Hello .... Your address is "
(base) PS C:\Users\LENOVO\Downloads\t23> .\example1.cmd Tiya
"Hello Tiya.... Your address is "
(base) PS C:\Users\LENOVO\Downloads\t23> .\example1.cmd Tiya Ulhasnagar
"Hello Tiya.... Your address is Ulhasnagar"
(base) PS C:\Users\LENOVO\Downloads\t23>
OUTLINE
TIMELINE
Ln 2, Col 38 Spaces: 4 UTF-8 CRLF Batch Signed out
26°C Sunny
Search
ENG US
13:49 27-01-2026

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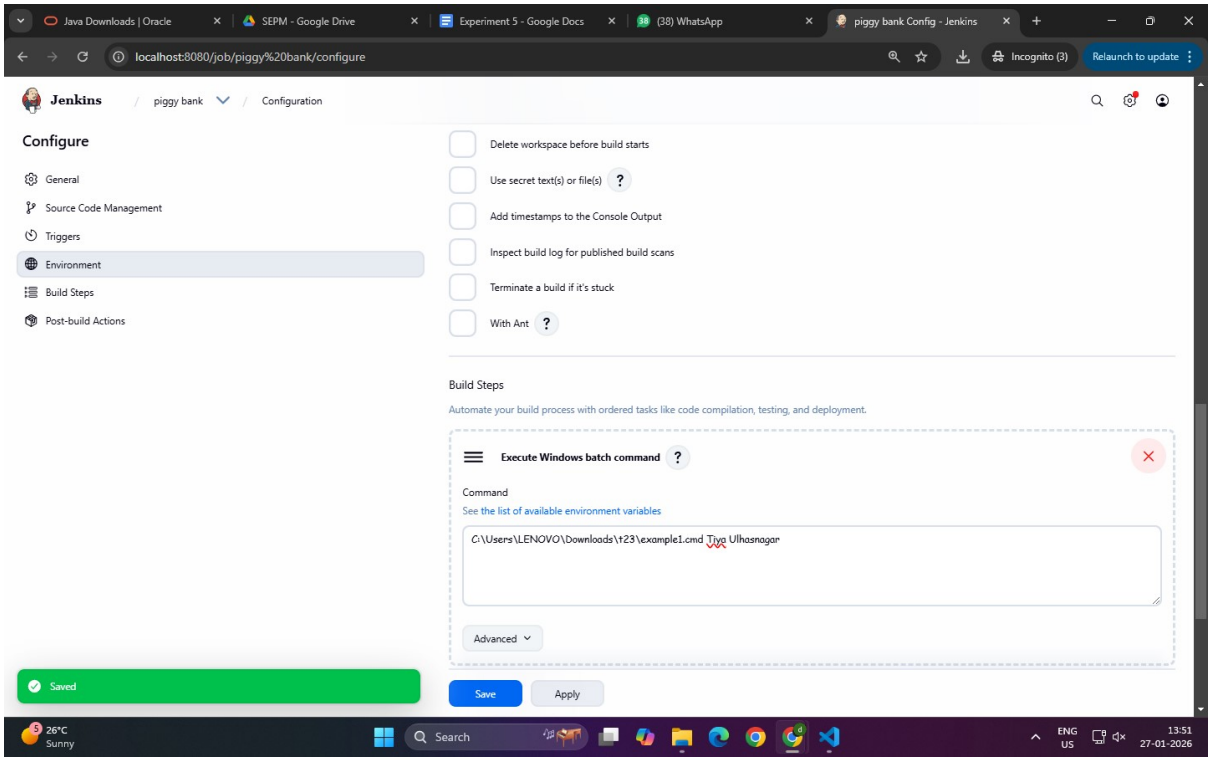

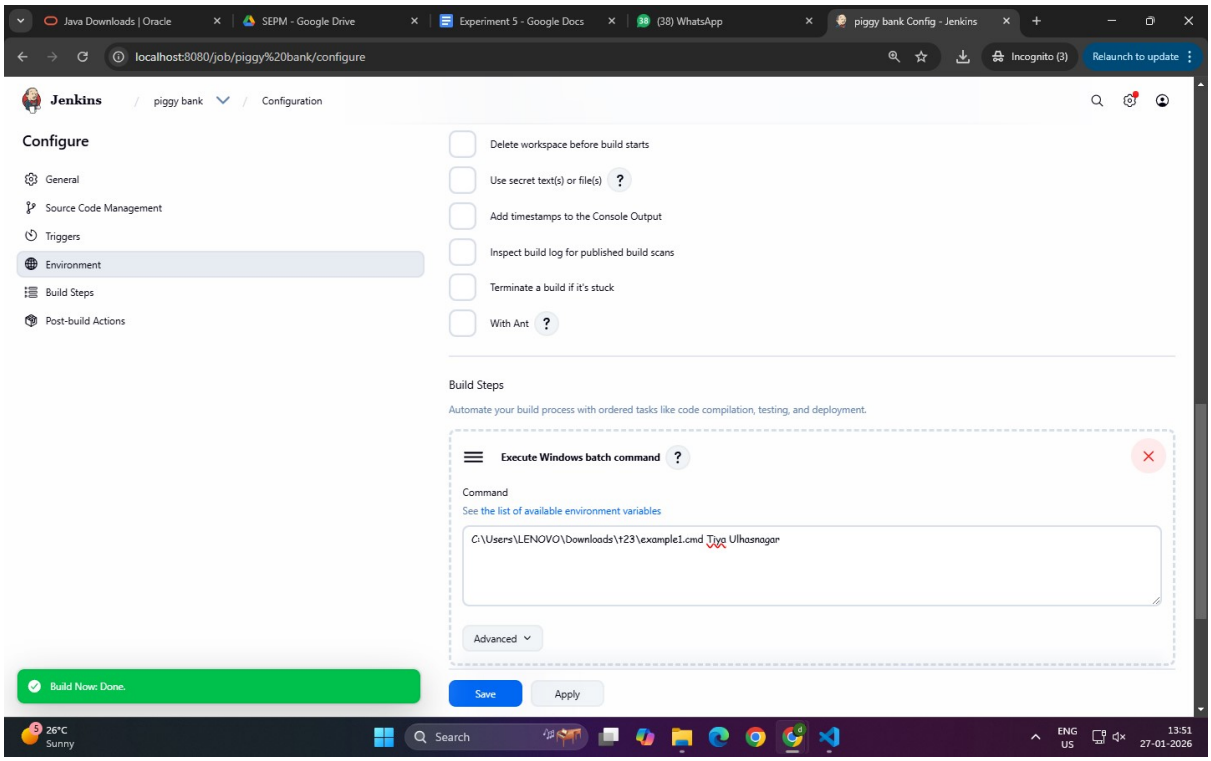

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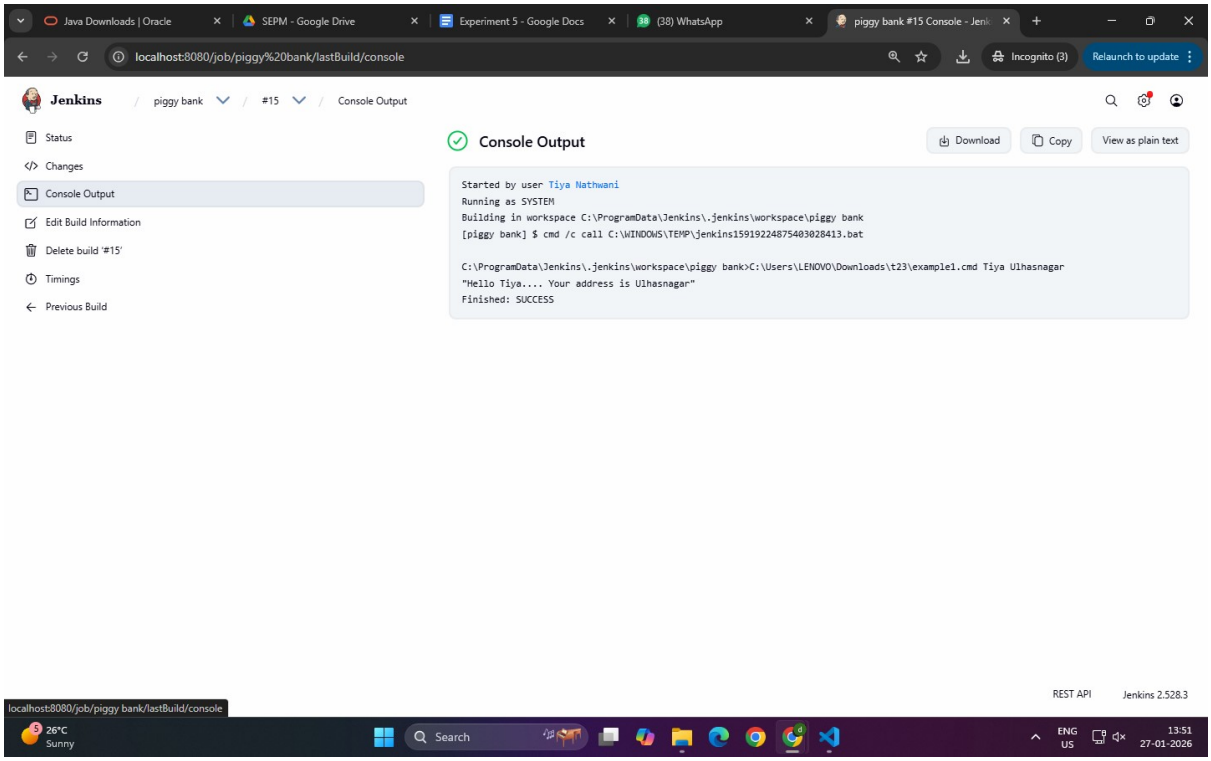

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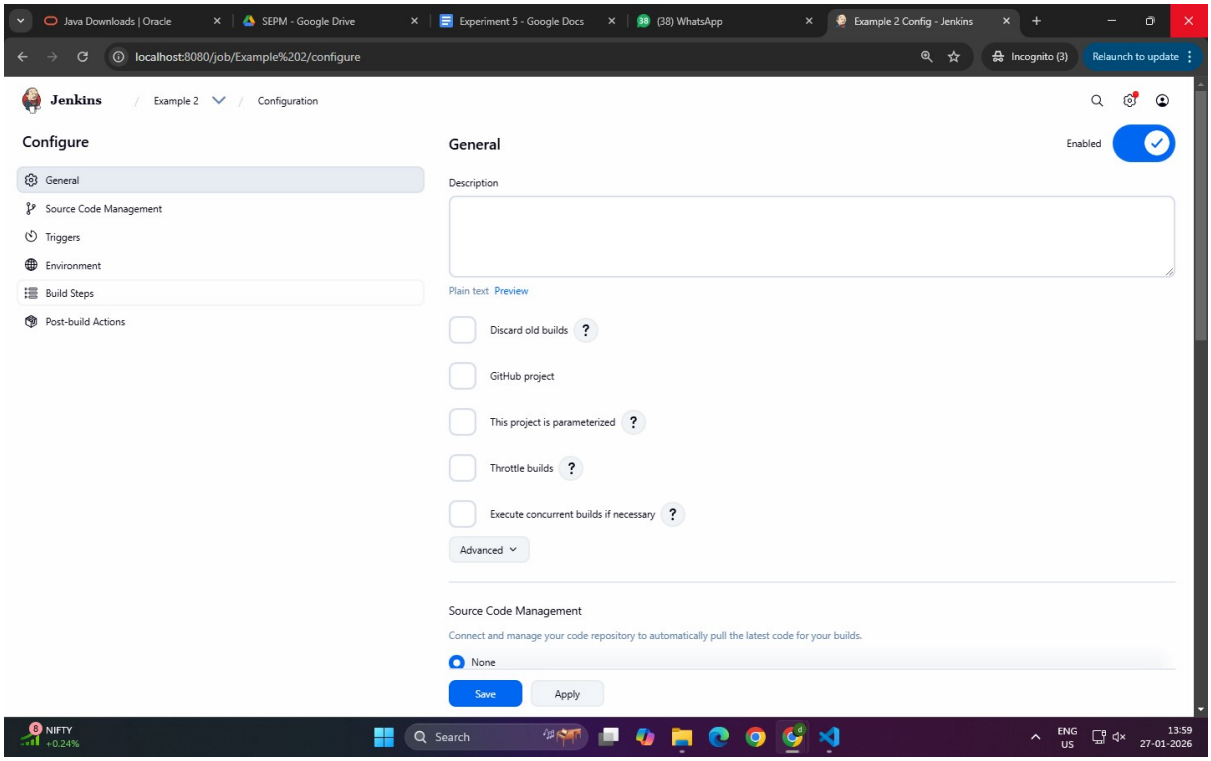
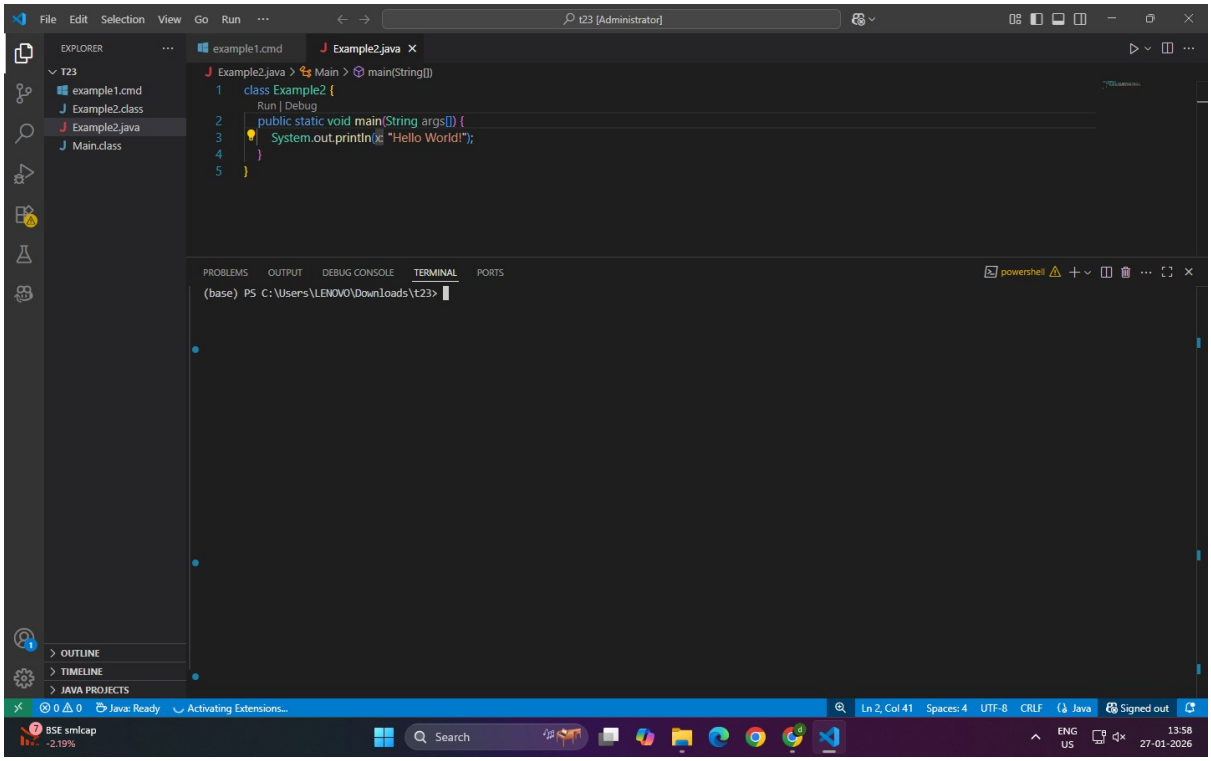
Example 2

Name: Arzoo Vaswani Batch: T23 Roll No. 125





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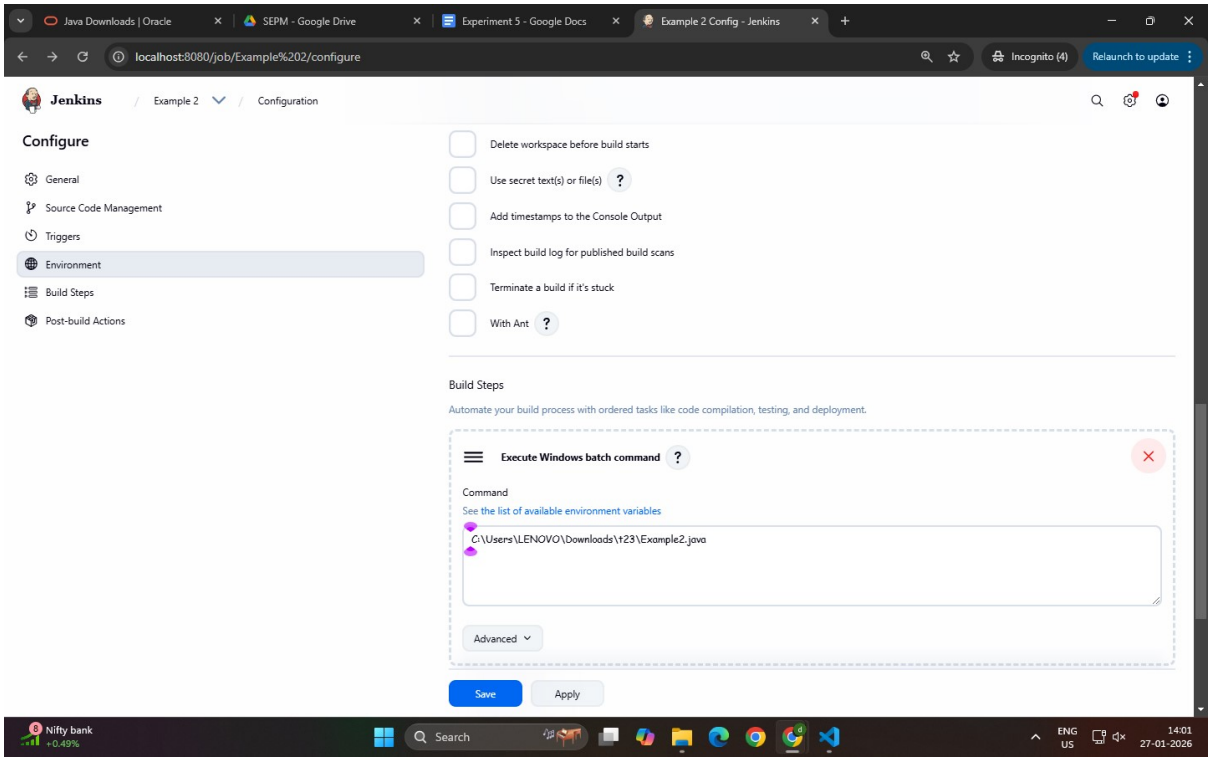

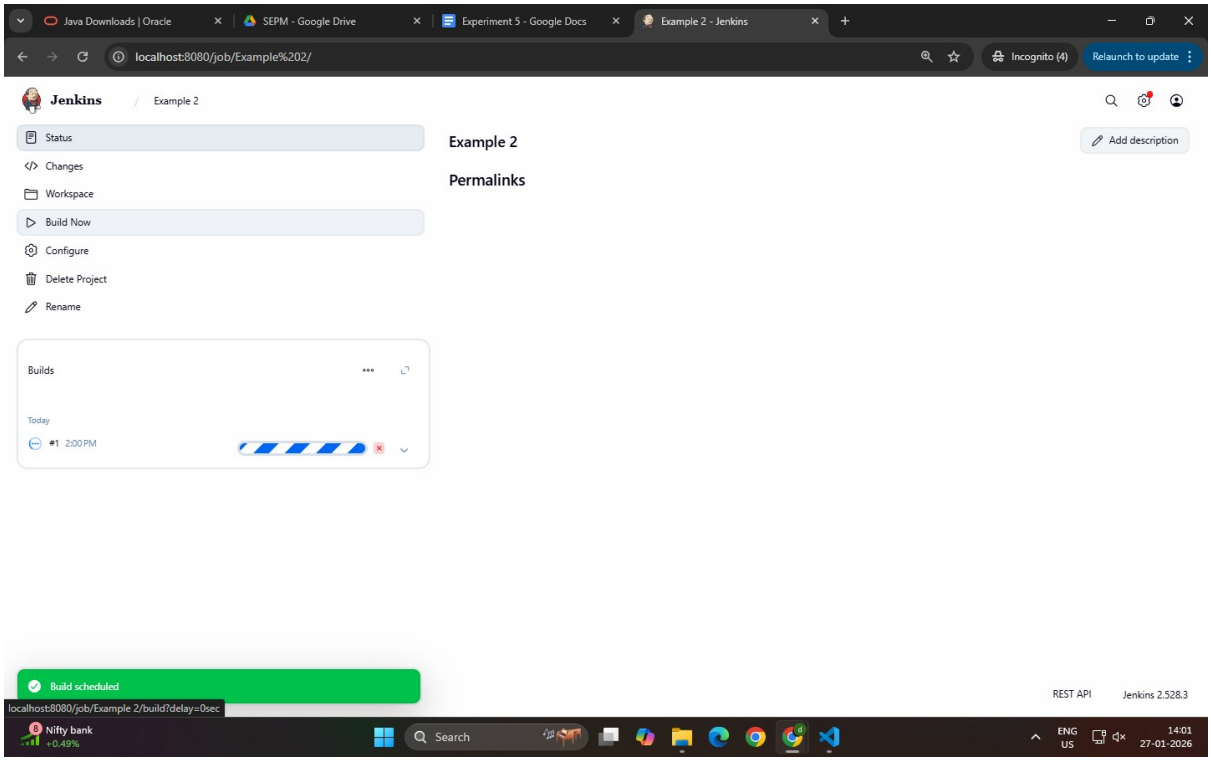

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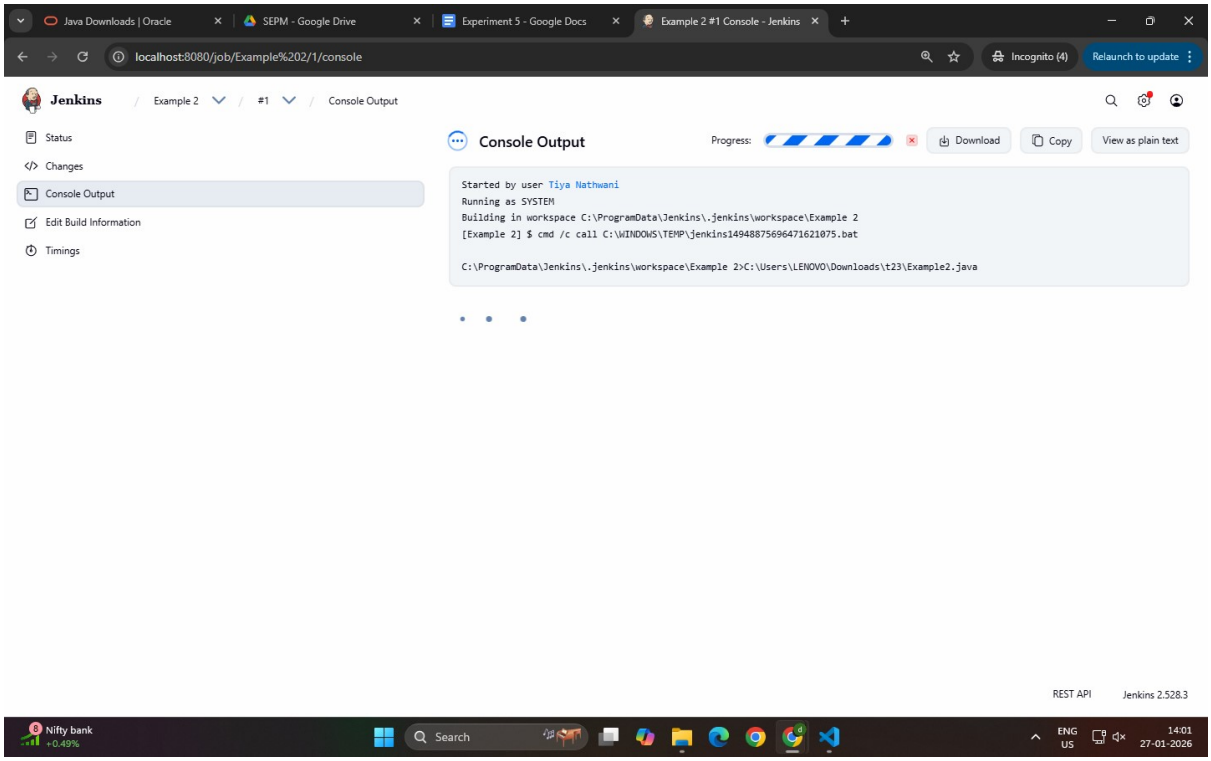
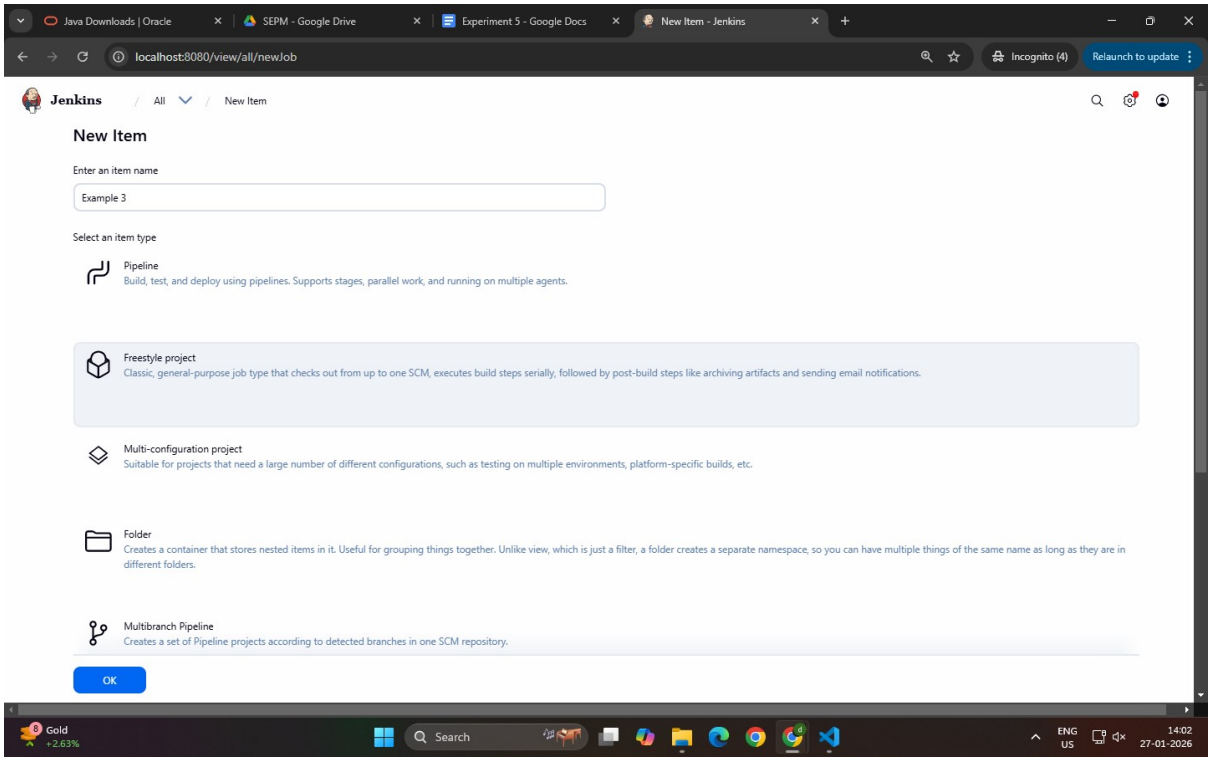
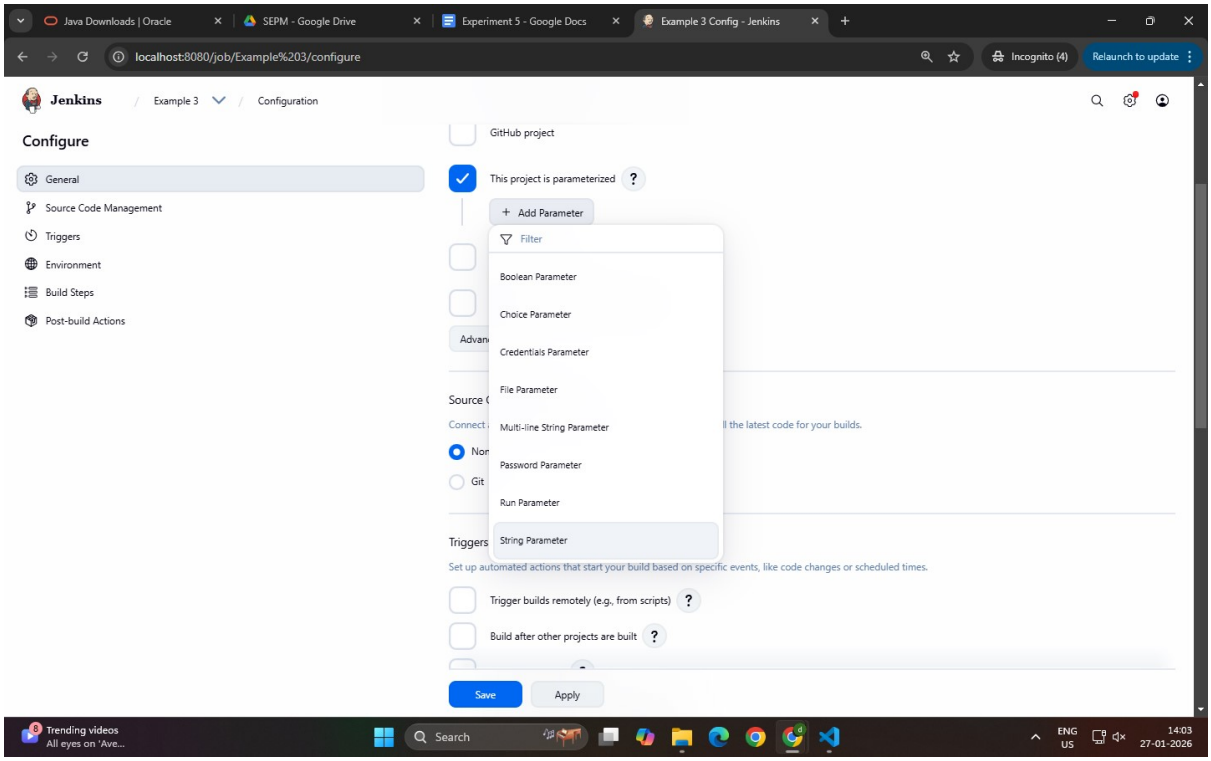
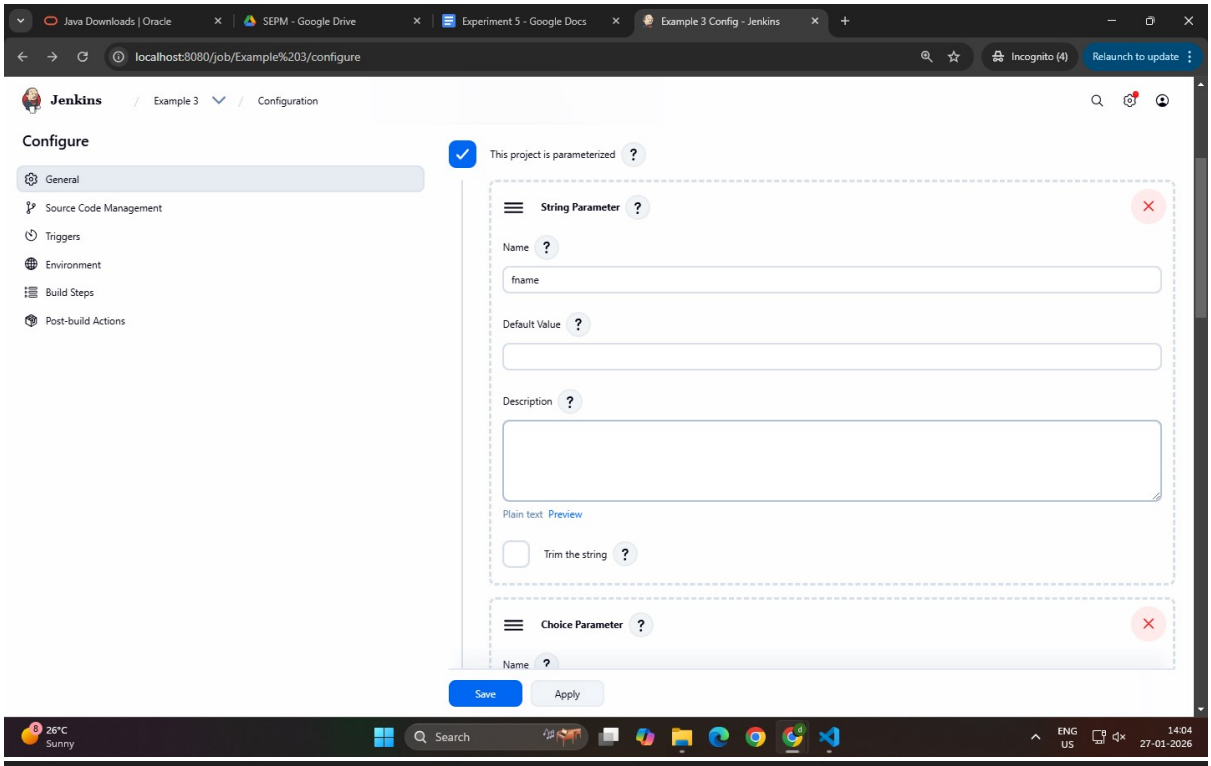
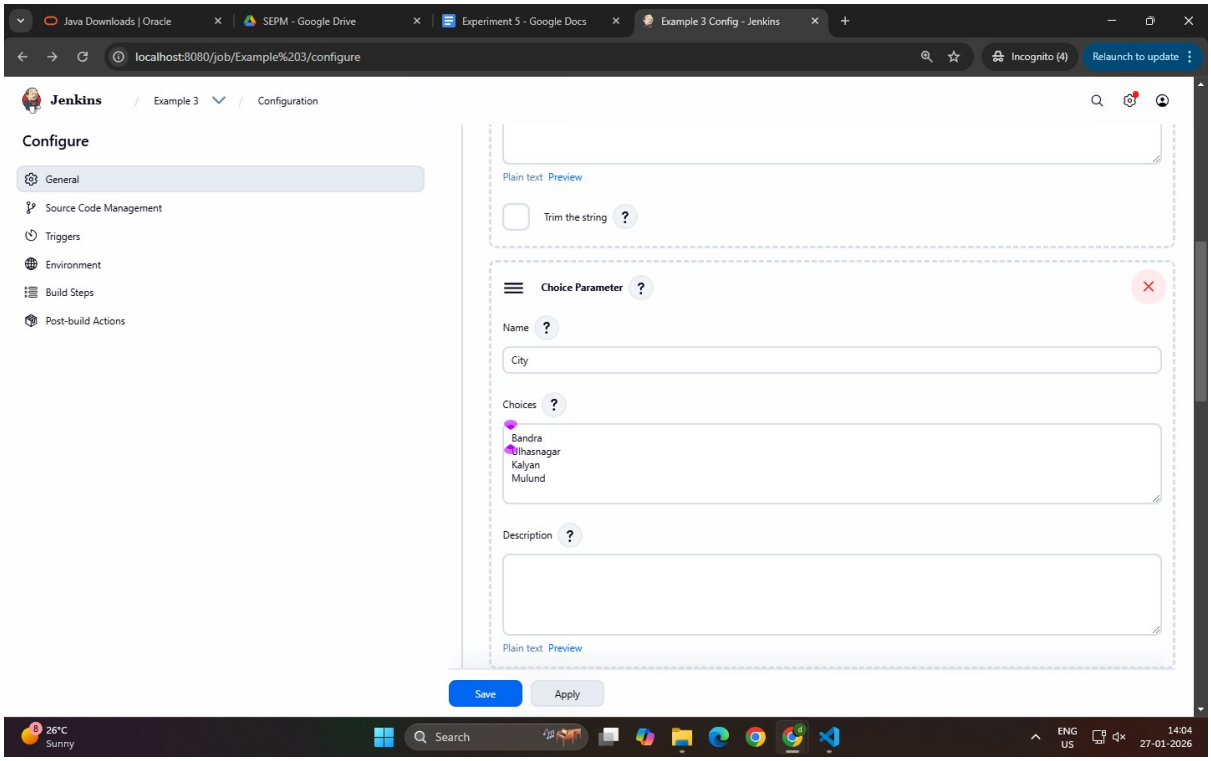


Example 3

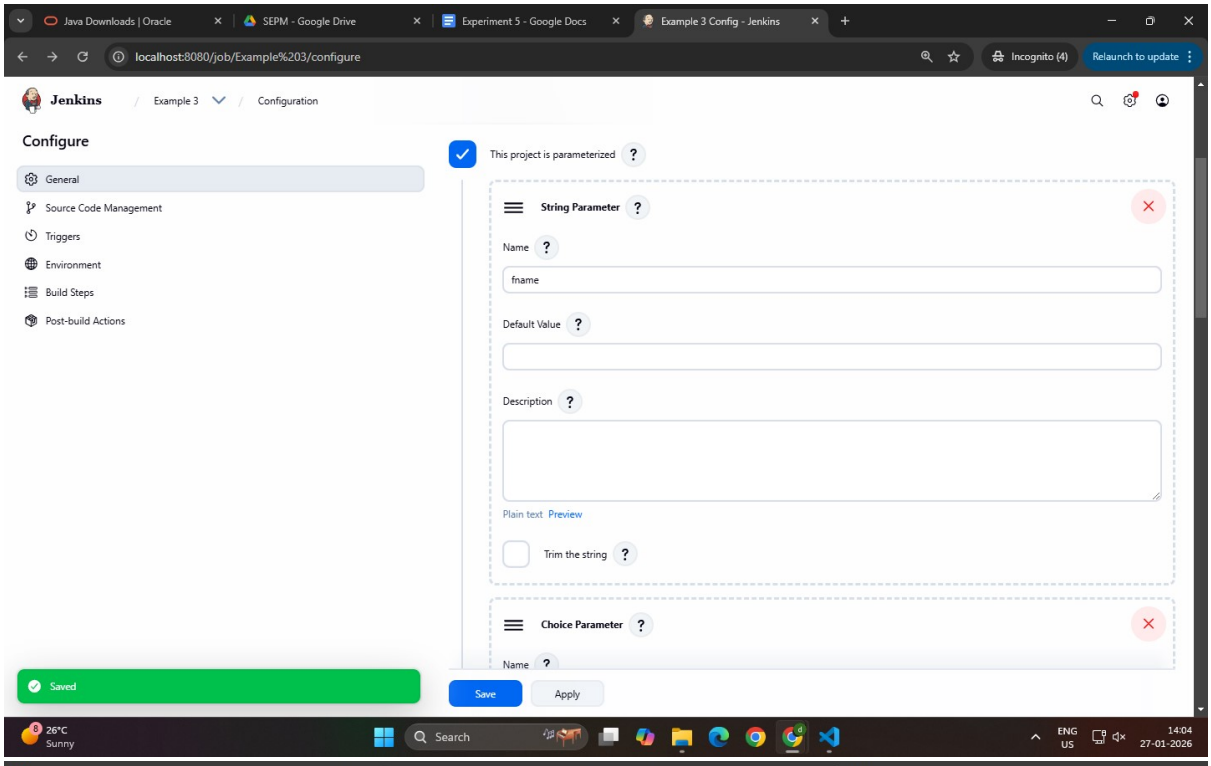

Java Downloads | Oracle
SEPM - Google Drive
Experiment 5 - Google Docs
New Item - Jenkins
localhost:8080/view/all/newJob
Jenkins
All
New Item
New Item
Enter an item name
Example 3
Select an item type
Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.
OK

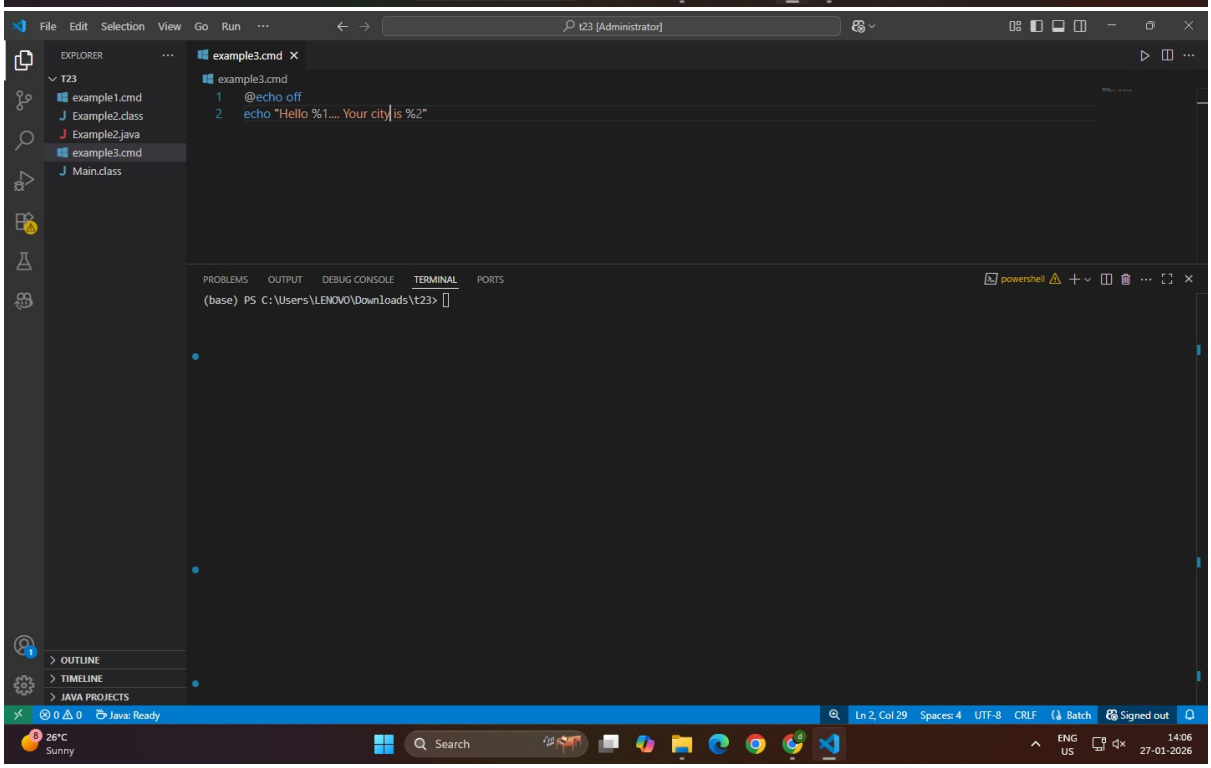

Java Downloads | Oracle
SEPM - Google Drive
Experiment 5 - Google Docs
Example 3 Config - Jenkins
localhost:8080/job/Example%203/configure
Jenkins
Example 3
Configuration
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
GitHub project
This project is parameterized
Add Parameter
Filter
Boolean Parameter
Choice Parameter
Credentials Parameter
File Parameter
Multi-line String Parameter
Password Parameter
Run Parameter
String Parameter
Triggers
Set up automated actions that start your build based on specific events, like code changes or scheduled times.
Trigger builds remotely (e.g., from scripts)
Build after other projects are built
Save
Apply


Java Downloads | Oracle
SEPM - Google Drive
Experiment 5 - Google Docs
Example 3 Config - Jenkins
localhost:8080/job/Example%203/configure
Incognito (4)
Relaunch to update
Jenkins
Example 3
Configuration
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
This project is parameterized
String Parameter
Name
fname
Default Value
Description
Plain text Preview
Trim the string
Choice Parameter
Name
Save
Apply
26°C
Sunny
Search
ENG
US
14:04
27-01-2026


Java Downloads | Oracle
SEPM - Google Drive
Experiment 5 - Google Docs
Example 3 Config - Jenkins
localhost:8080/job/Example%203/configure
Incognito (4)
Relaunch to update
Jenkins
Example 3
Configuration
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Plain text Preview
Trim the string
Choice Parameter
Name
City
Choices
Bandra
Ulhasnagar
Kalyan
Mulund
Description
Plain text Preview
Save
Apply
26°C
Sunny
Search
ENG
US
14:04
27-01-2026

Name: Arzoo Vaswani Batch: T23 Roll No. 125





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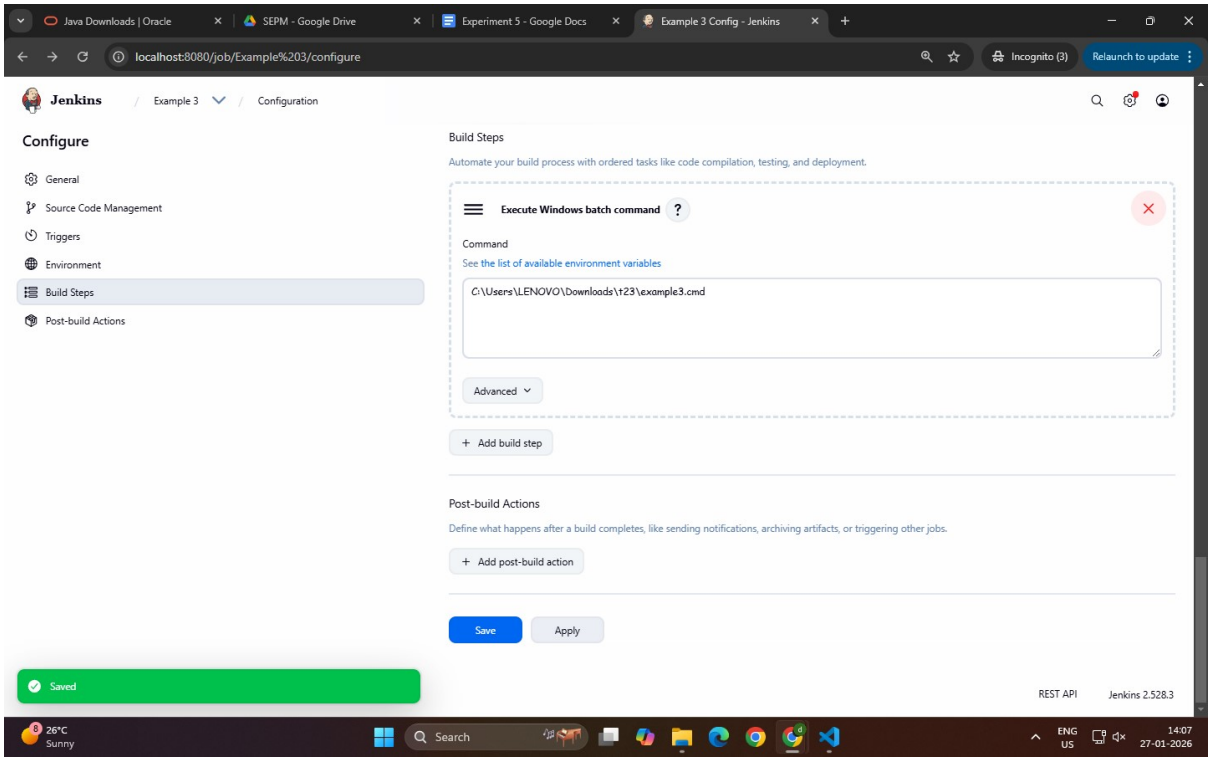

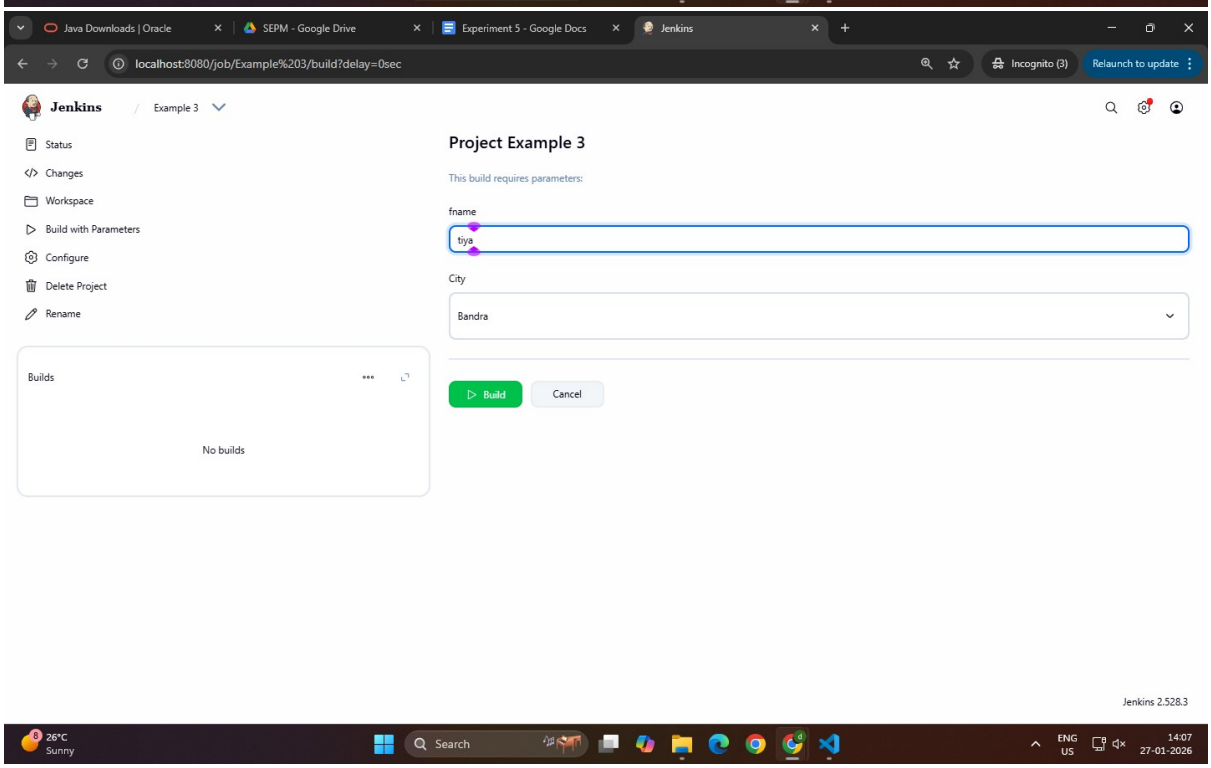

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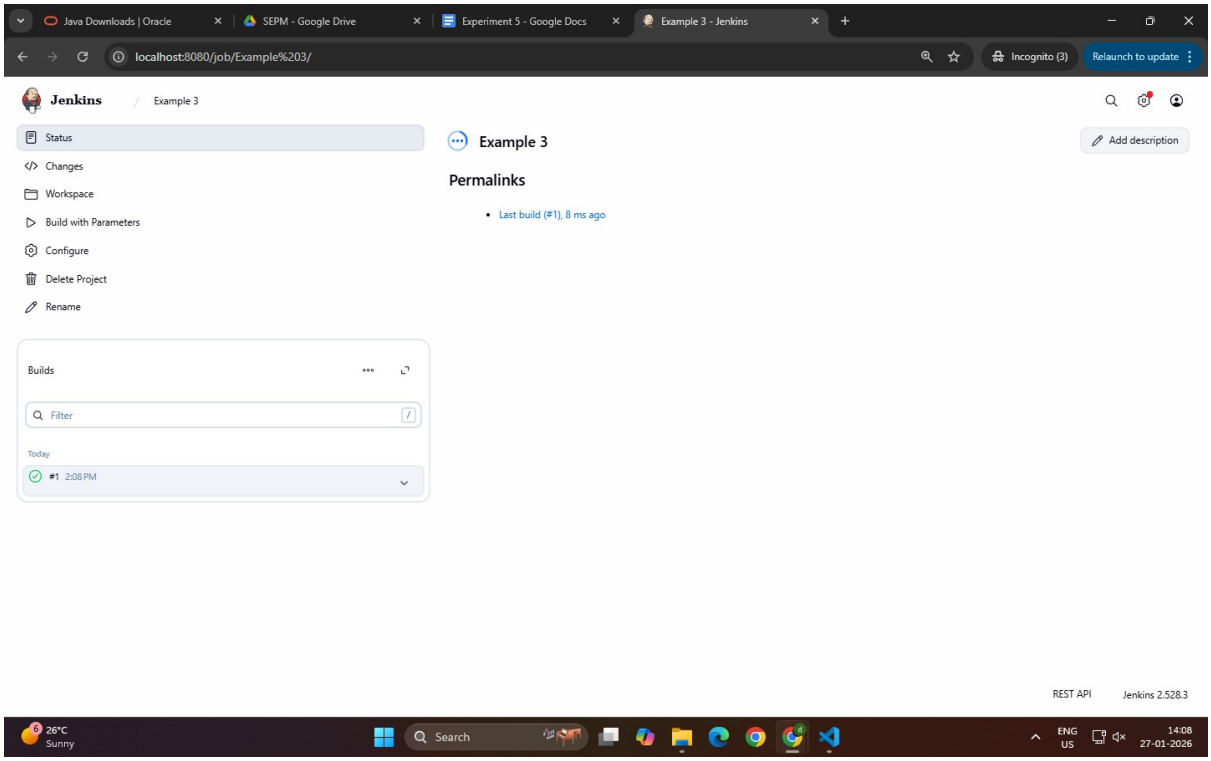
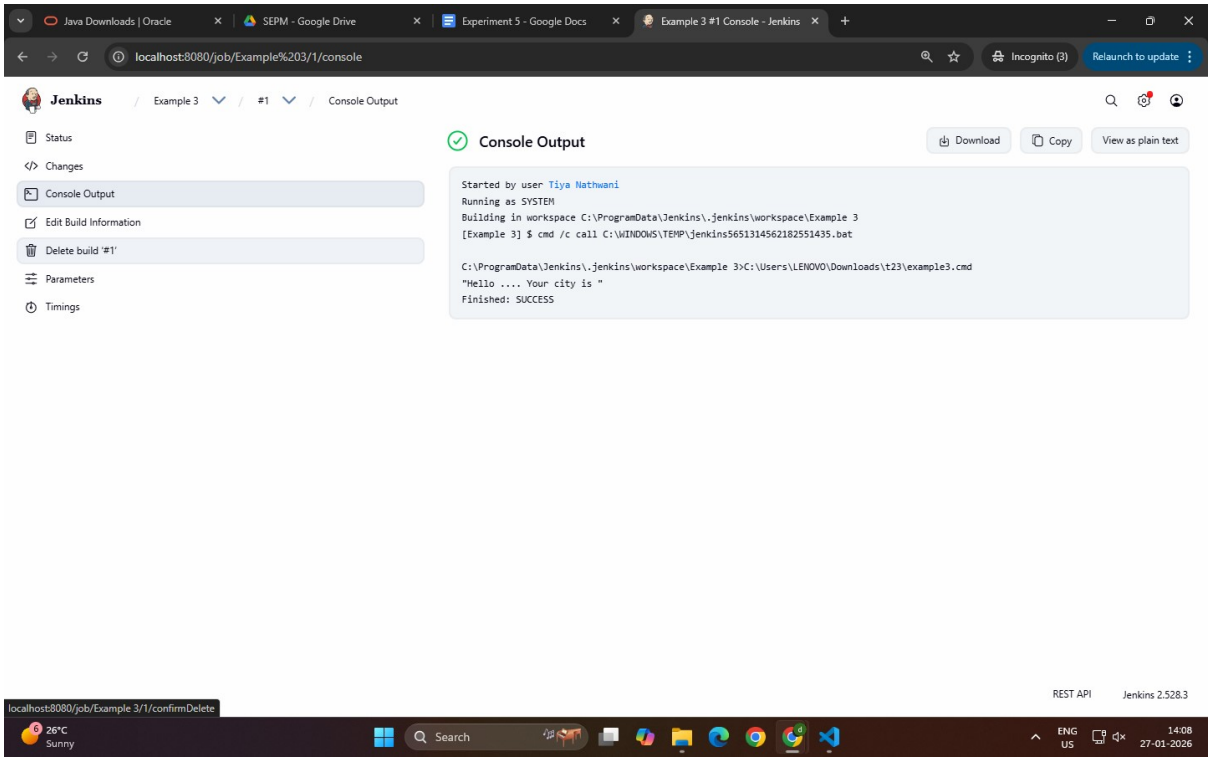
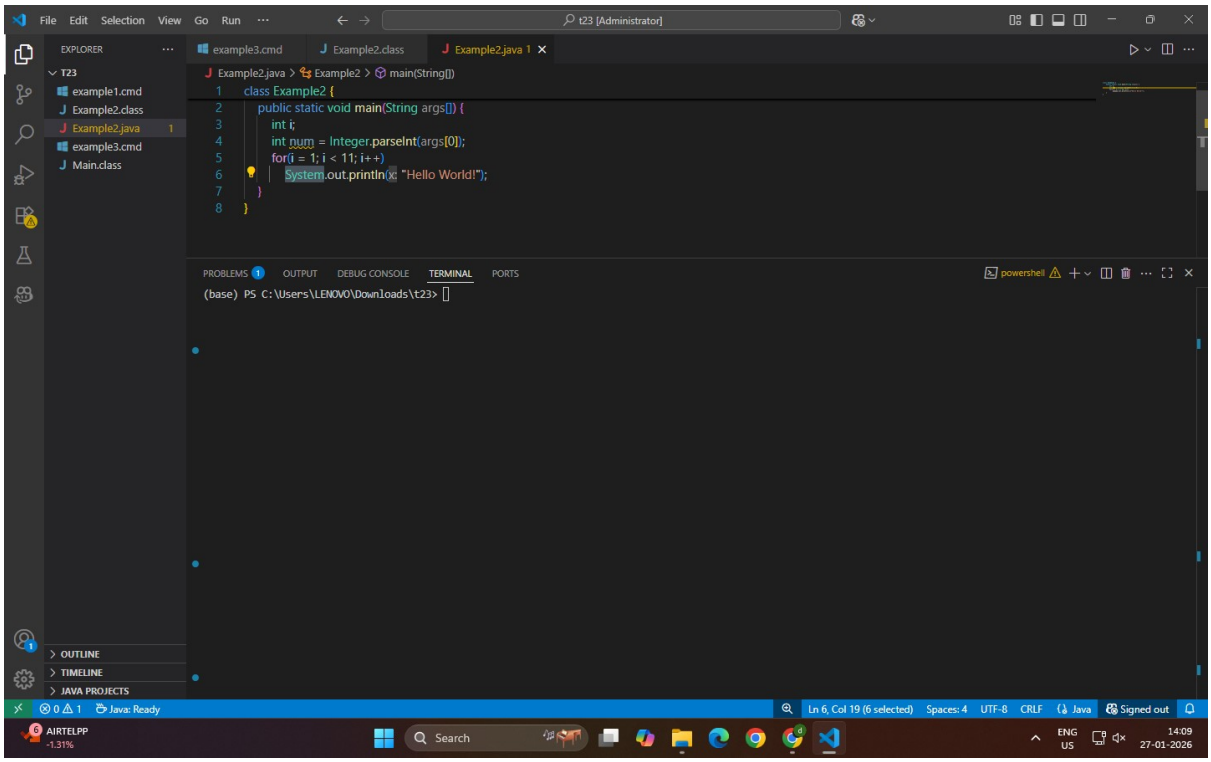
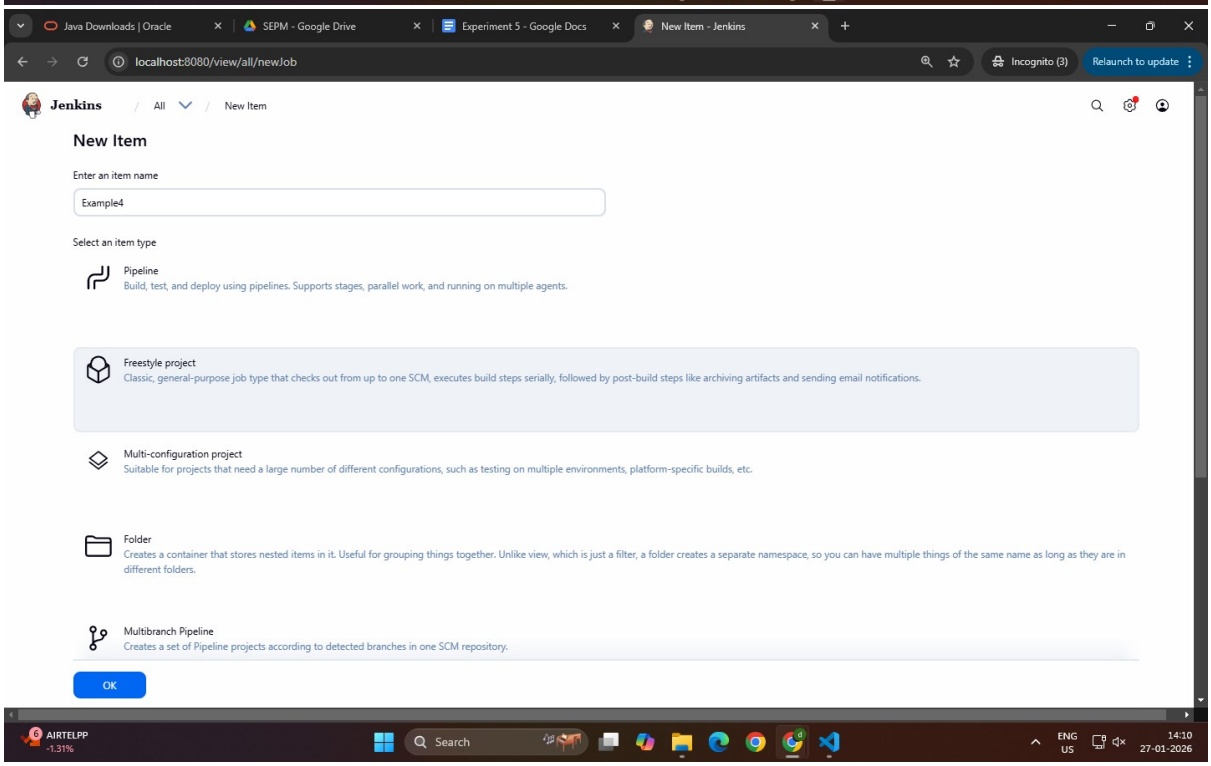




3.2


Example2.java > Example2 > main(String[])
class Example2 {
public static void main(String args[]) {
int i;
int num = Integer.parseInt(args[0]);
for(i = 1; i < 11; i++)
System.out.println(x: "Hello World!");
}
}
(base) PS C:\Users\LENOVO\Downloads\t23>


Jenkins
New Item
Enter an item name
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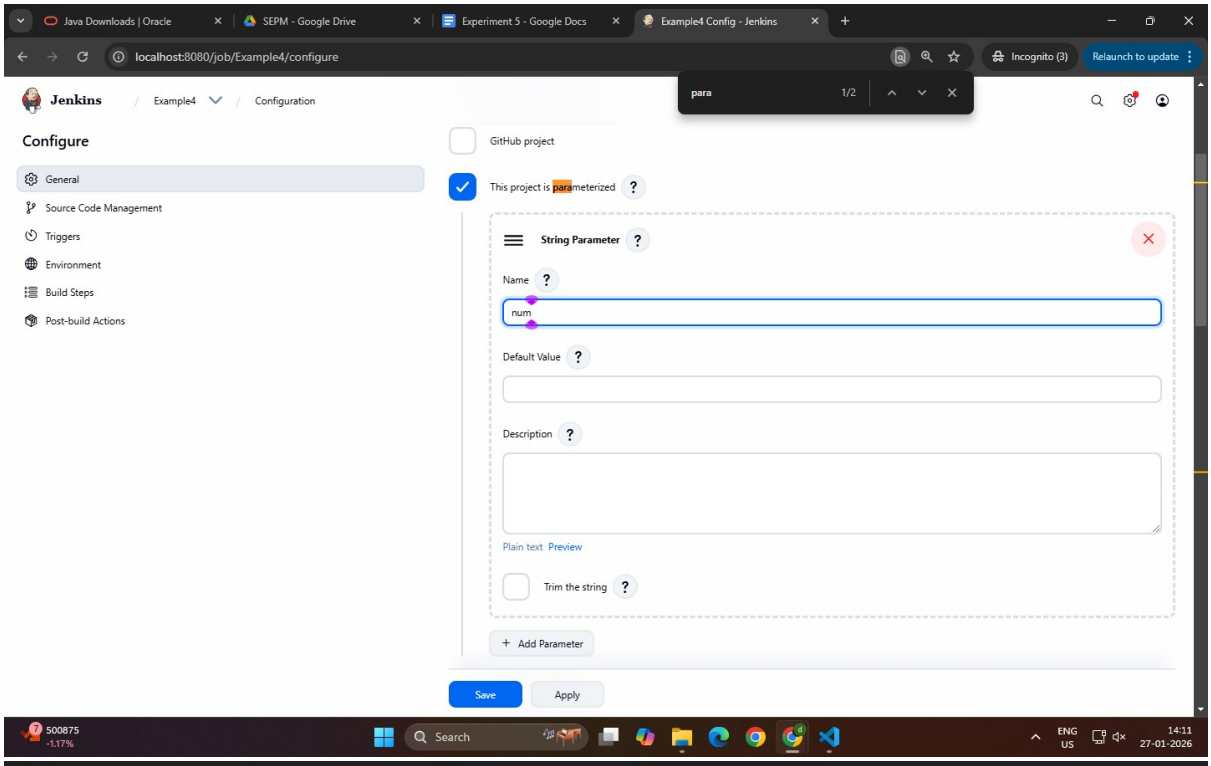
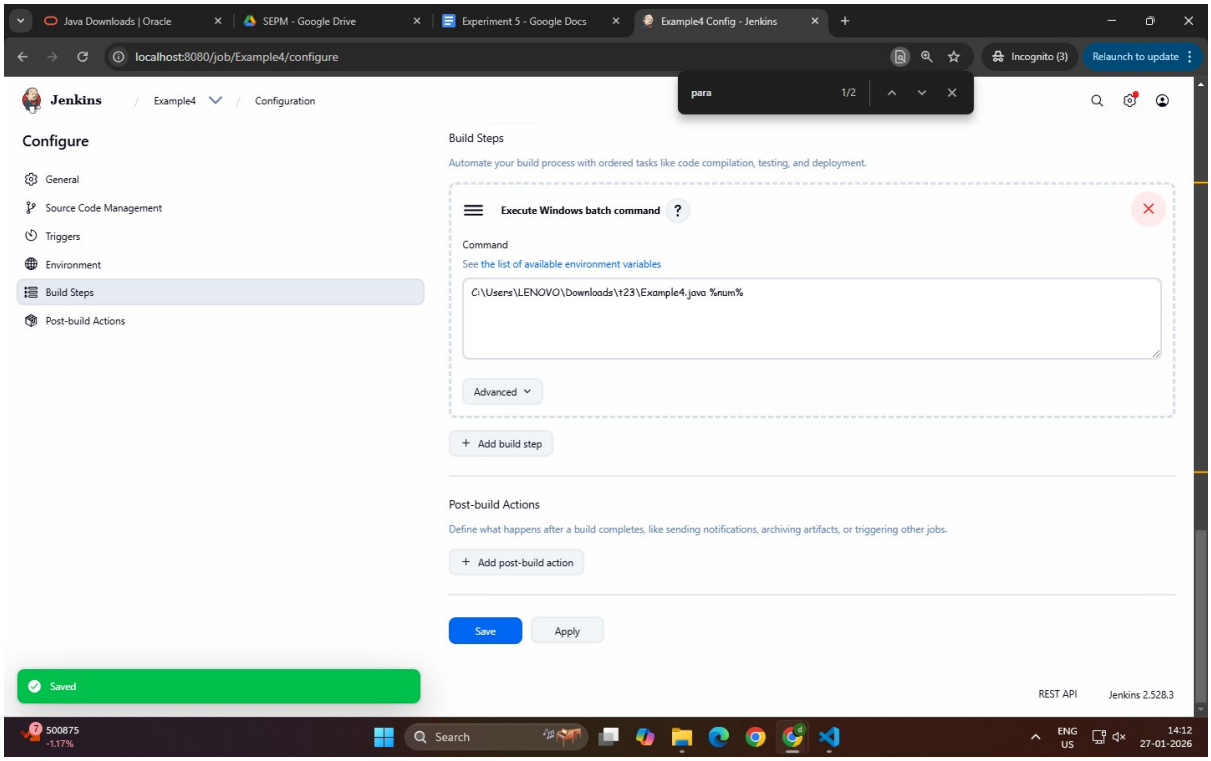
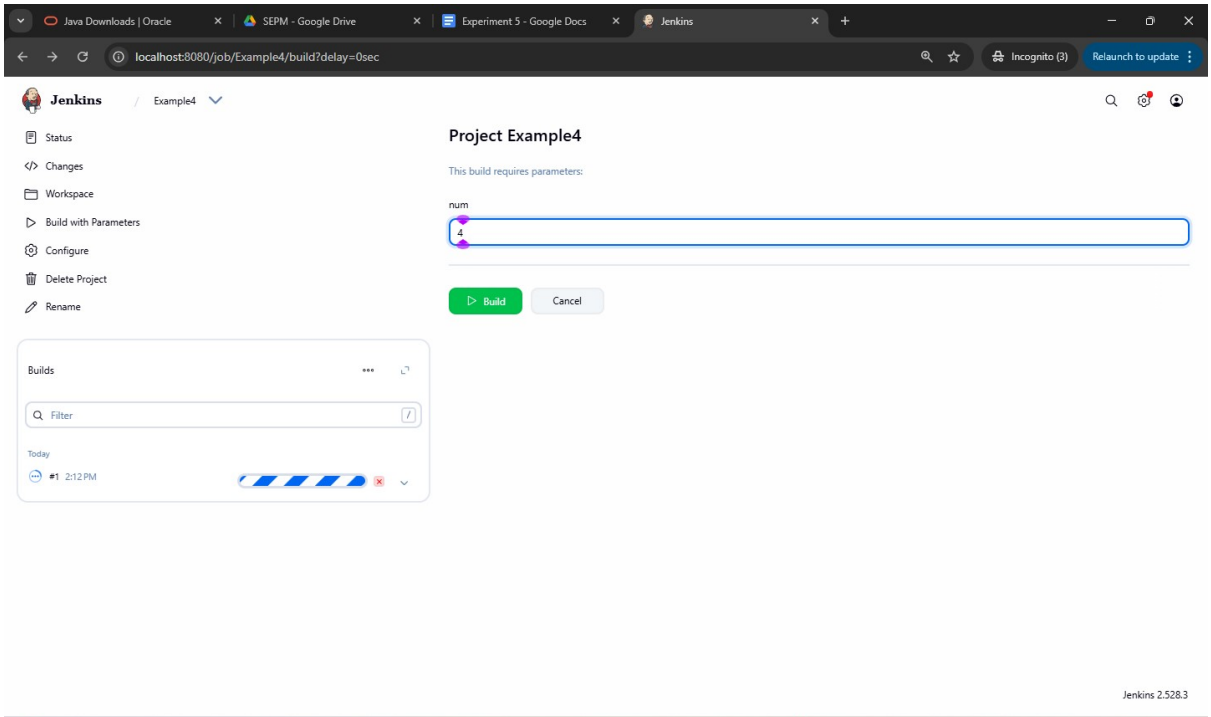
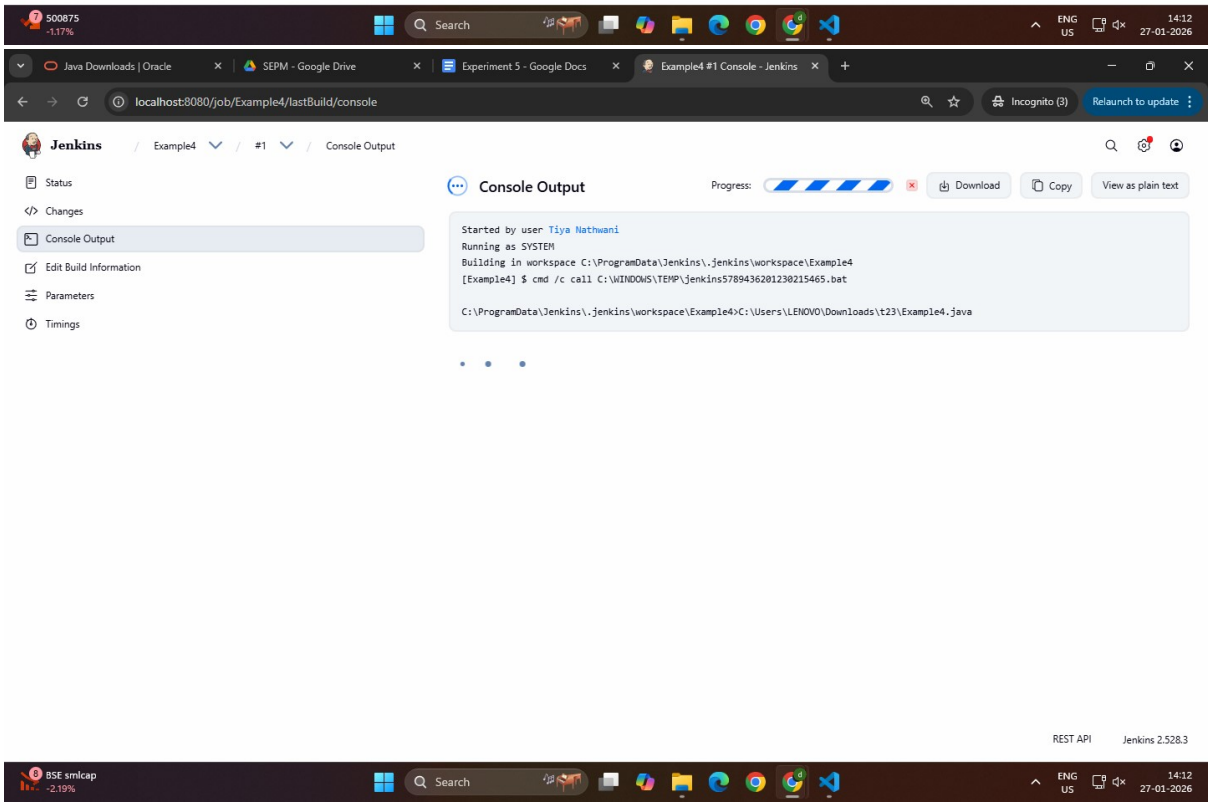

Java Downloads | Oracle
SEPM - Google Drive
Experiment 5 - Google Docs
Example4 Config - Jenkins
localhost:8080/job/Example4/configure
Incognito (3)
Relaunch to update
Jenkins
Example4
Configuration
para
1/2
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
GitHub project
This project is parameterized
String Parameter
Name
num
Default Value
Description
Plain text Preview
Trim the string
Add Parameter
Save
Apply
Search
ENG
US
14:11
27-01-2026


Java Downloads | Oracle
SEPM - Google Drive
Experiment 5 - Google Docs
Example4 Config - Jenkins
localhost:8080/job/Example4/configure
Incognito (3)
Relaunch to update
Jenkins
Example4
Configuration
para
1/2
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Build Steps
Automate your build process with ordered tasks like code compilation, testing, and deployment.
Execute Windows batch command
Command
See the list of available environment variables
C:\Users\LENOVO\Downloads\t23\Example4.java %num%
Advanced
Add build step
Post-build Actions
Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.
Add post-build action
Save
Apply
Saved
REST API
Jenkins 2.528.3
Search
ENG
US
14:12
27-01-2026


Java Downloads | Oracle
SEPM - Google Drive
Experiment 5 - Google Docs
Jenkins
localhost:8080/job/Example4/build?delay=0sec
Incognito (3)
Relaunch to update
Jenkins
Example4
Status
Changes
Workspace
Build with Parameters
Configure
Delete Project
Rename
Project Example4
This build requires parameters:
num
4
Build
Cancel
Builds
Filter
Today
#1 2:12 PM
Jenkins 2.528.3


Java Downloads | Oracle
SEPM - Google Drive
Experiment 5 - Google Docs
Example4 #1 Console - Jenkins
localhost:8080/job/Example4/lastBuild/console
Incognito (3)
Relaunch to update
Jenkins
Example4
#1
Console Output
Status
Changes
Console Output
Edit Build Information
Parameters
Timings
Console Output
Progress:
Download
Copy
View as plain text
Started by user Tiya Nathwani
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\.jenkins\workspace\Example4
[Example4] $ cmd /c call C:\WINDOWS\TEMP\jenkins5789436201230215465.bat
C:\ProgramData\Jenkins\.jenkins\workspace\Example4>C:\Users\LENOVO\Downloads\t23\Example4.java
REST API
Jenkins 2.528.3